

Atoms in density functional theory

Assignment #1

Electron Eigenvalues with VASP

Calculate the electronic energy levels of the He, C, N and O atoms.

Comment on the number of electrons and energy levels obtained by the calculation, in comparison with the atomic number of each atom. Attribute to each level the appropriate quantum state (s,p,d,...).

Discuss how the results change by modifying the size of the cell (hint: draw a graph to show how the occupied energy level change by varying the unit cell).

- « Use a cubic cell containing one atom (POSCAR file). Put the atom in the center of the cell.
- « Set the POTCAR file using the appropriate GGA potential.
- « Use the default KPOINTS file with one k point (Γ -point only).
- « Initialize the wave functions with random numbers (set ISTART=0 in the INCAR).
- » In the OUTCAR file search for "band energies" to get the energy and occupation of the individual orbitals.
- » Check your results for different cell sizes (5 Å, 7 Å, 9 Å, and 11 Å).

Eigenvalues in the spin polarized setup

Repeat the calculations for the eigenvalues of He, C, N and O atoms, in the spin polarized setup.

Compare with results from the spin unpolarized calculations. Are the results obtained by the spin unpolarized setup meaningful? Motivate your answer, mentioning also which systems could turn to be more suitable for the two setups, respectively.

- « Set ISPIN=2 in the INCAR file.
- « Use the 9 Å large cell.
- » Mind that in the OUTCAR file you have two sets of orbital (*i.e.*, for spin up and down, separately).

Eigenvalues and Symmetry

Calculate the electronic energy levels of the O atom, in the spin polarized setup, in cell with reduced symmetry.

Compare with the results obtained by using the cubic cell: What change can you observe? Discuss the results considering the underlying quantum-mechanical principle.

- « Use a cell with reduced symmetry (*e.g.*, $8 \text{ \AA} \times 9 \text{ \AA} \times 10 \text{ \AA}$). Keep the atom in the center of the cell.

Spatial distribution of electrons

Visualize the spatial distribution of selected orbitals of the O atom, in the spin polarized setup. Extract a partial electron density by performing a non-self consistent calculation (reading the electron density and wave functions previously calculated in a self consistent calculation). Select an energy interval which includes only one specific orbital.

- « Set LPARD=.TRUE. in the INCAR file to activate the partial density calculation.
- « Set EINT= *min max* in the INCAR file, (where *min* and *max* define the energy range containing any electronic state). Alternatively, you can set IBAND= *n*, which select the *n*-th energy level of both spin up and spin down electrons, simultaneously.
- « Read the electronic charge density and wave functions obtained from the self consistent calculation: copy the CHGCAR and WAVECAR files from the self-consistent calculation folder and set ICHARG=11 and ISTART=1 in the INCAR file.
- « Avoid to overwrite the CHGCAR and WAVECAR by setting LCHARG=.FALSE. and LWAVE=.FALSE. in the INCAR file.
- » Use VESTA to visualize the selected state
Open VESTA and open (read in) the file PARCHG

(Optional) Quantitative look at the spatial distribution

Compare the spatial distribution of the s and p states (separately) of the O atom, along the (100) (*i.e.*, *x* axis), (010) (*i.e.*, *y* axis), (001) (*i.e.*, *z* axis) and (110) directions.

- » Use VESTA to extract the spatial distribution of the electronic charge along a line: go to Utilities > Line Profile, select the initial and final coordinates of the segment, select the number of data points.
- » Compare graphically the various profile lines extracted by VESTA.